The Price of Agreement: Gravity as the Geometry of Consensus

Bernhard Mueller

Pragma Research Inc.

Chiang Mai
Thailand

Corresponding author: Bernhard Mueller, bernhard@floatingpragma.ai

Submission date: September 11, 2026
Public revision: October 25, 2026

Essay written for the Gravity Research Foundation 2027 Awards for Essays on Gravitation

Abstract. To a trained physicist this essay's starting point may sound absurd: it assumes no spacetime. It starts from finite observers repairing disagreements and asks how their records could support a geometric description. Read-after-write dependence supplies a finite causal order, a specified repair rule has a slow band of rank three, and declared reference-preserving repairs supply entropy identities and constrained equilibria. These are finite ingredients of a conditional route to Jacobson's equation of state. Physical area, boost, stress, fixed-volume generalized-entropy stationarity and a common continuum refinement remain additional premises. The finite algebra is checked in Lean and exact code; the complete physical antecedent is unproved [7, 8].

A small thing to fix

Quantum field theory describes three of the four forces to a precision that embarrasses the instruments, and it describes the fourth not at all. Gravity is weak, it is universal, and it is carried, if it is carried by anything, by a massless particle of spin two. The obvious repair is to write that particle down on the flat spacetime everything else lives on and let the machinery run.

The difficulties are instructive. A massless spin-two field that couples to energy must couple to its own energy, and Deser's consistent self-coupling construction yields general relativity [10]. Perturbatively quantized pure Einstein gravity has a two-loop divergence requiring a new counterterm [12]. Weinberg and Witten obstruct an emergent graviton carrying energy in a Lorentz-covariant theory with a covariant conserved stress tensor [11]. These results have precise hypotheses; they do not exclude every quantum-gravity route. They motivate the question pursued here: could the geometry needed to describe locality, causality and fields arise from the observers' records themselves?

The proposal is to begin before the geometric stage has been assigned. Finite observers have bounded memory and a few ports each, and must agree with their neighbours about what they jointly see. Long-known arguments then offer a route from their records to geometry and, under further physical premises, Einstein's equation. The order of those arguments matters, and so do the joints between them.

Three accessories not in the box

The air above your kitchen table has a horizon. Pick a point in it and a direction, and imagine an observer accelerating through that point in that direction for long enough. A surface forms behind it that no signal will cross in its direction, a Rindler horizon, and Unruh showed in 1976 that the observer reads the vacuum beyond that surface as warm, at a temperature proportional to the acceleration [1]. In 1995 Ted Jacobson took this seriously in every direction at once [2]. Give each such horizon an entropy proportional to its area and

require that the heat crossing it obey Clausius, $\delta Q = T\,dS$, at every point and along every direction. The geometry then has no freedom left. It has to curve in the one way that keeps the accounts straight, and that curvature is $G_{ab}+\Lambda g_{ab} = 8\pi G\,T_{ab}$, which is Einstein's equation arrived at by accountancy. In 2016 Jacobson expressed the equilibrium premise, under his geometric and matter assumptions, as stationarity of small-ball entanglement entropy at fixed volume [3]. Verlinde and Padmanabhan later read the same thermodynamics into Newton's law and into a wide class of gravity theories [4, 5].

The derivation works perfectly provided you have bought three accessories that are not in the box. Nothing in it says why a horizon's entropy should count area, why a surface that is an accident of one observer's acceleration should have a temperature, or why the state of the world should be one in which entropy is stationary on every small region in every direction at all times. Jacobson calls the last of these a hypothesis, which is the polite word. Eling, Guedens, and Jacobson worked out what happens when it fails: the equation of state acquires dissipative terms and Einstein's equation stops being exact [6]. The equilibrium carries the whole weight of the argument, and nothing in the argument holds the world in it. The derivation reasons about a clock, a thermometer, and an equilibrium that nobody has located on the board.

The claim of this essay is that a network of finite observers supplies concrete candidates for parts of all three. Boundary counts offer area proxies, faithful states carry modular flows, and specified repair laws realize constrained entropy equilibria. A cut count is not automatically physical entropy, a modular parameter is not automatically an accelerating thermometer's clock, and finite equilibrium is not fixed-volume entanglement stationarity. The Observer Patch Holography programme checks the finite mathematics and states the remaining physical joints [7, 8].

What an observer is

An observer here holds a bounded amount of state, reads the outside through finitely many

ports, writes records that later moves may not touch, and shares boundaries with neighbours whose records must agree with its own where they overlap, because two accounts of one shared fact cannot both stand. The programme assumes three things [7]. Observers form a federation on a spherical support, each with twelve ports at the vertices of an icosahedron, joined to neighbours through seams, a seam being two ports routed to each other. Observers agree: any datum two of them jointly read means the same to both, on every overlap. Whatever agreement leaves unconstrained is maximally random. The third assumption is Jaynes's maximum-entropy principle [13] promoted from advice to statisticians to a law of nature.

The finite model starts with this spherical wiring, rather than a physical spacetime metric. In its strict-descent repair realization, accepted moves lower a mismatch potential and terminate on a finite network. On a federation of 81,920 patches, sixteen tested asynchronous schedules land on the same public normal form [8]. That is a finite agreement result. Continuing stochastic repairs and coherent clocks have their own specified laws; termination does not by itself produce their physical equilibrium.

Where before and after come from

With no global timeline, "before" can only mean dependence. A record written after reading another record depends on it, and the transitive closure of authenticated read-after-write dependence is a strict partial order on records. Lamport saw in 1978 that this is the only time a distributed system has [14]; he was thinking about computers that had to agree on the order of things without being able to see one another, which is the universe's problem with better documentation. In the observer network the relation is machine-checked to be irreflexive, transitive, and locally finite [7], and a locally finite partial order is the object Bombelli, Lee, Meyer, and Sorkin proposed in 1987 as the discrete precursor of spacetime [15], and in a distinguishing spacetime the causal order fixes the metric up to a conformal factor, with a volume element fixing the rest [16, 17].

Two records that do not depend on each other are incomparable in the order. Calling that relation physical spacelike separation requires an order-faithful spacetime identification. The log alone supplies no shared present. It does supply a height, the length of the longest chain of dependencies ending at a record, and Lean proves that height strictly increases along the order and that a chain attaining it exists. Height is an ordinal; a reading of proper time requires a clock calibration.

Where three plus one comes from

The spatial directions come from the ports. Twelve ports on an icosahedron carry the rotation group A_5, and the twelve-dimensional space of port readings splits as

$$P_{12} \;\cong\; \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \tag{1}$$

an exact decomposition with no repeated summand. Repair acts on this split as a low-pass filter. The uniform seam-mean rule is $T = I - L/60$ with L the icosahedral Laplacian, so each band fades at its own rate per step, $(55 + \sqrt{5})/60$ for one triplet, $9/10$ for the quintet, and $(55 - \sqrt{5})/60$ for the other triplet. Content that has faded before a second observer can compare it anchors nothing, so only the slowest band lasts, and its rank is three. The normalized response kernel converges to $4P_{\text{slow}}$, a projector of rank three, and the integer records embed densely in the three-dimensional Euclidean space it defines [7]. A struck bell ends on one note for the same reason: the other modes rang down first.

Adjoin a real axis to that space and write $Q(t, x) = t^2 - g(x, x)$. It has one positive and three negative directions, and source-unit spatial directions label its future null rays. Placing source heights and spatial readbacks in this carrier requires a time scale and causal compatibility; the signature alone does not identify a physical spacetime. A separate theorem selects the Lorentz cone within its stated nonzero closed convex pointed class under rotation and boost invariance, with a time orientation picking the future half [9]. Those invariances are premises about the candidate influences and readouts.

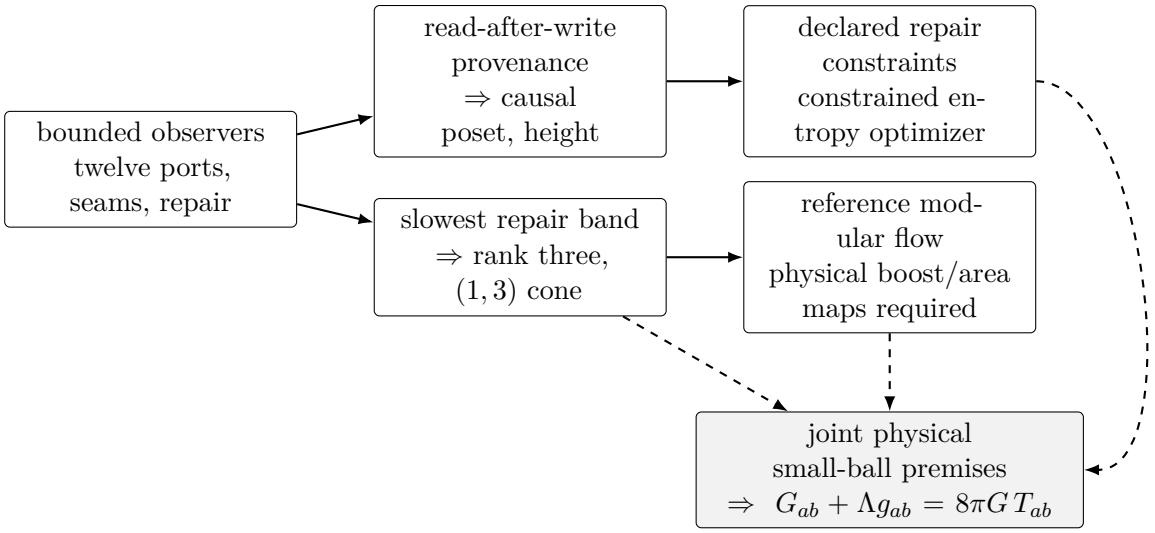


Figure 1: Finite ingredients and a conditional field equation. Dashed arrows require the joint physical entropy, area, boost, stress, volume and refinement premises; the finite witnesses do not supply them all.

On a supplied population of q^3 golden-ratio sites with a complete-neighbour read law, the provenance order is proved equal to a layered order with an exact cone sandwich, and finite runs read a Myrheim–Meyer dimension of 4.09 at $q = 55$ against 3.02 and 2.00 for two- and one-dimensional control populations, and an interval ordering fraction of 0.093 against the flat four-dimensional value 1/10 [9].

What finite entropy stationarity supplies

Take a declared faithful reference π and a repair that keeps the settled fibre labels while resampling within each fibre from that reference. This conditional projection is an idempotent recovery map [18]; it preserves π and contracts relative entropy, $D(pP \,\|\, \pi) \le D(p \,\|\, \pi)$. With the stated forward/reverse path convention and support conditions, the finite fluctuation identity is $\langle e^{-\sigma} \rangle = 1$ and its mean-production inequality follows from Jensen's inequality [7]. The reference, conserved data and transition law are part of the construction.

A fixed point has the reference's conditional distribution within each fibre. At fixed fibre probabilities it minimizes relative entropy to that reference, by the finite Pythagorean identity proved in Lean. For a counting reference this is the ordinary maximum-entropy

statement; other declared moment constraints give the corresponding Gibbs family. Stationarity holds along variations preserving those constraints. This is a precise finite equilibrium, but Jacobson's premise concerns physical generalized entropy under state and shape variations at fixed physical volume. Neither a conserved occupation nor a fixed list of patches supplies that premise.

A faithful finite state supplies a modular flow with generator $K = -\log \rho$ [19]. Bisognano and Wichmann identify vacuum-wedge modular flow with a Lorentz boost under their field-theory hypotheses, with normalization $K = 2\pi B$ [20]. For its faithful reference τ, the finite cap construction begins with a declared split $K = -\log \tau = 2\pi B + Z$ and proves the exact first law $S(p) - S(\tau) = \langle K \rangle_p - \langle K \rangle_\tau - D(p\|\tau)$. The central term Z supplies its own edge pairing. Identifying B with a physical boost, the modular parameter with the relevant clock, and the edge pairing with physical boundary entropy are distinct obligations. Its interpretation as a physical Rindler horizon requires those maps.

Boundary counting needs no gravity. On specified populations and read laws, crossing counts provide geometric area proxies. Finite state algebras also have exact bulk and edge entropy splits. These are different statements: graph symmetry alone does not make entropy proportional to the number of cut seams. If the physical identification gives Bekenstein's $S = A/(4G)$ [21], it fixes $G = A/(4S)$ in natural units. Establishing that same area and entropy dictionary on the common physical history remains necessary.

The equation

With these physical identifications and the stationarity premise supplied, the rest is the small-ball calculation. Fix a small ball of radius ℓ around an event in the assumed smooth geometry and admit variations at fixed physical volume. Assume its generalized entropy has the form

$$S_{\text{gen}} = S_{\text{bulk}} + \frac{A}{4G}, \tag{2}$$

and that its first variation vanishes about the specified vacuum reference. The physical

ball-modular identification gives the leading term $\delta S_{\mathrm{bulk}} = (8\pi^2\ell^4/15)\,\delta\langle T_{00}\rangle$, while fixed-volume geometry gives $\delta A = -(4\pi\ell^4/15)\,\delta G_{00}$. With the common remainder bounds, frame compatibility and vacuum-reference identification required by the theorem, cancellation in the shrinking-ball limit yields

$$G_{ab}\,k^a k^b = 8\pi G\,T_{ab}\,k^a k^b \tag{3}$$

for every null k. Under the stated entropy and modular normalizations, the factor 8π follows from the ball coefficients without a fit; the physical scale G remains to be identified. Null values determine a symmetric tensor up to a multiple of the metric, a finite linear-algebra fact proved in Lean. The contracted Bianchi identity and stress conservation force that multiple to be constant on a connected region. The full equation $G_{ab} + \Lambda g_{ab} = 8\pi G\,T_{ab}$ follows on the stated C^3 Lorentzian domain once the tensors, null balance and Ward identity are supplied [7, 9].

A Lorentz-invariant vacuum contribution is proportional to the metric, and $g_{ab}k^a k^b = 0$ along null rays. Null stress data therefore leave that contribution undetermined. This does not forbid other local measurements or calculations, or by itself solve the vacuum-energy problem. A separate fixed-capacity branch proposes $\Lambda\,\ell_P^2 = 3\pi/N$. For specified candidates with N near 3.3×10^{122}, this gives values within one percent of the Planck central coordinate $1.09\times10^{-52}\,\mathrm{m}^{-2}$ [7]; the corresponding DESI DR2 display puts those candidates 2.64 and 2.81 standard deviations low. These are conditional comparisons, not a source-selected cosmological constant. The branch also predicts $w = -1$ under its cosmological identification, so an established departure would reject that branch [23].

What gravity is

The proposed interpretation is that gravity describes how an observer network's effective geometry responds to matter. The conditional theorem makes that proposal precise once the entropy, stress and continuum interfaces are present [7]. A persistent repair pattern can

have a recurring cost, but this does not identify a physical rest mass or equate that cost with gravitational stress. Those identifications require the relevant field response, kinetic normalization, clock and common matter action. MICROSCOPE's agreement of inertial and gravitational response at the level of one part in 10^{15} [22] is a physical target for the construction, not a measurement of its repair cost.

Agreement alone does not make gravity inevitable. The finite order, rank-three band, modular flow and constrained optimizer are constructive ingredients. They do not imply null balance on every physical small ball. The equation becomes compulsory when the joint physical premises hold; showing that one continuing repair history realizes them, including a responsive geometry, is the remaining task. A fixed-background wave description cannot supply that response merely by having Lorentzian signature.

What is proved and what is identified

The finite results include the partial order, height, band split, rank-three Gram identity, $(1, 3)$ carrier, first-law and fluctuation identities, declared Gibbs optimizers, cap Clausius inequality and null tomography. Lean checks these algebraic statements and the conditional tensor completion; exact computational receipts and analytic limits retain their own scopes [8]. Physical use requires event and clock identification, geometric modular normalization, independent stress and curvature, area and volume calibration, and fixed-volume generalized-entropy stationarity with controlled errors on one common refinement family [7, 9]. These are several obligations, not one change of units. The finite witnesses discharge parts of them; the definition of an observer does not discharge the full bundle.

Four buildings

Jaynes in 1957, Lamport in 1978, Bombelli, Lee, Meyer, and Sorkin in 1987, and Jacobson in 1995 supplied the pieces. The test is whether one observer history realizes the physical equilibrium that makes the geometric equation follow. Jacobson compared quantizing that equation with quantizing sound. An equation of state needs its equilibrium; completing the

proposed physical identification would make gravity the price of agreement.

References

[1] W. G. Unruh, "Notes on black-hole evaporation," *Phys. Rev. D* **14**, 870–892 (1976).

[2] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260–1263 (1995).

[3] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116**, 201101 (2016).

[4] E. Verlinde, "On the origin of gravity and the laws of Newton," *J. High Energy Phys.* **2011**(4), 29 (2011).

[5] T. Padmanabhan, "Thermodynamical aspects of gravity: new insights," *Rep. Prog. Phys.* **73**, 046901 (2010).

[6] C. Eling, R. Guedens, and T. Jacobson, "Nonequilibrium thermodynamics of spacetime," *Phys. Rev. Lett.* **96**, 121301 (2006).

[7] B. Mueller, A. Osika, M. Poneder, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, "Finite observer consensus as a reconstruction principle: normal forms, the Standard Model Lie type, and a route to the Einstein field equation," PhilPapers record MUEFOC (2026). `https://philpapers.org/rec/MUEFOC`

[8] FloatingPragma, *Observer Patch Holography: Lean library, exact code receipts, finite certificates, and scientific claim registries* (2026). `https://github.com/FloatingPragma/observer-patch-holography`

[9] B. Mueller, A. Osika, M. Poneder, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, "Recovering observer spacetime and Einstein dynamics from overlap consistency," companion paper in [8], directory `paper/` (2026).

[10] S. Deser, "Self-interaction and gauge invariance," *Gen. Relativ. Gravit.* **1**, 9–18 (1970).

[11] S. Weinberg and E. Witten, "Limits on massless particles," *Phys. Lett. B* **96**, 59–62 (1980).

[12] M. H. Goroff and A. Sagnotti, "The ultraviolet behavior of Einstein gravity," *Nucl. Phys. B* **266**, 709–736 (1986).

[13] E. T. Jaynes, "Information theory and statistical mechanics," *Phys. Rev.* **106**, 620–630 (1957).

[14] L. Lamport, "Time, clocks, and the ordering of events in a distributed system," *Commun. ACM* **21**(7), 558–565 (1978).

[15] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-time as a causal set," *Phys. Rev. Lett.* **59**, 521–524 (1987).

[16] D. B. Malament, "The class of continuous timelike curves determines the topology of spacetime," *J. Math. Phys.* **18**, 1399–1404 (1977).

[17] S. W. Hawking, A. R. King, and P. J. McCarthy, "A new topology for curved space–time which incorporates the causal, differential, and conformal structures," *J. Math. Phys.* **17**, 174–181 (1976).

[18] D. Petz, "Sufficiency of channels over von Neumann algebras," *Q. J. Math. Oxford* **39**, 97–108 (1988).

[19] M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Mathematics 128 (Springer, Berlin, 1970).

[20] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17**, 303–321 (1976).

[21] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333–2346 (1973).

[22] P. Touboul et al. (MICROSCOPE Collaboration), "MICROSCOPE mission: Final results of the test of the equivalence principle," *Phys. Rev. Lett.* **129**, 121102 (2022).

[23] DESI Collaboration, "DESI DR2 results II: Measurements of baryon acoustic oscillations and cosmological constraints," arXiv:2503.14738 (2025).